MIL-W-48855C (AR)
AMENDMENT 3
December 17, 1987
SUPERSEDING
AMENDMENT 2
12 September 1986

MILITARY SPECIFICATION

WARHEAD BODY ASSEMBLY
FOR
WARHEAD SECTION, GM, HE, M258

This amendment forms a part of Military Specification MIL-W-48855C (AR), dated 17 May 1985, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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* 2.1.1 Specifications and standards.

Add the following after "STANDARDS":

"FEDERAL

FED. - STD-151-Metals; Test Methods"

* 2.1.2 Other Government documents, drawings and publications.
Add the following after "9344462 -Warhead Body Assembly":

"9297072 -Preform, Powder Metal"

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3.4 add the following requirement:

"Isolation with warhead case and P3-9/P3-2 shorted". ≥ 1meg ohm

* Delete paragraph 3.5 in its entirety.

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* Add new paragraph 3.5 as follows:

"3.5 Process control document. The contractor shall supply a process control document which is to include descriptions of all production processes, procedures and equipment (including in-process inspection equipment) used to manufacture and inspect the production material. These descriptions shall identify the basic fabrication process, fabrication equipment (including methods for evaluating status of equipment), controls associated with the fabrication processes and in-process inspections. All initial and revised process control documents shall be submitted to the technical office for review and approval of completeness prior to its use in production (see 6.4). It should be noted that it is not the government's intention to approve the contractor's process itself, but merely the completeness and adequacy of the process control document."

* 3.7 after "free from seams, injurious segregations, cracks" add the following: ", porosity"

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* 4.4.1 Inspection lot formation: Delete "Additionally, preform lot size shall be limited to the quantity sintered in a single heat and for batch process, shall be limited to a single batch." and substitute "Additionally, preform lot size shall be limited to the quantity sintered in single oven at one time, and for a batch process, shall be limited to a single batch."

* 4.4.2 Examination. See MIL-A-48078.

b. Sample size for mechanical and hardness tests.

In table labeled "SAMPLE SIZE" delete column labeled "DENSITY**" and substitute the following:

"DENSITY**
10% of lot/
Minimum
10% of lot
10% of lot
10% of lot

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* 4.4.2 Examination. See MIL-A-48078.

c. Reduced sampling plan.

In table labeled "SAMPLE SIZE" delete column labeled "DENSITY**" and substitute the following:

"DENSITY**
5% of lot/
minimum
5% of lot"

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4.4.2.2 add new major categories in the appropriate columns as follows:

"Category Major	Exam or Test	AQL or 100%	Req. Para.	Para Ref/ Insp. Meth.
109	Presence of threads in connector center pin	100%	3.1	Gage
110	Torque test of electrical plug connector."	100%	3.1	Gage

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* 4.4.2.4 add new major category in the appropriate columns as follows:

"Category Major	Exam or test	AQL or 100%	Req. Para.	Para Ref/ Insp. Meth.
105	Chemical analysis (PMP)	2/	3.1	4.5.10"

* Add new note 2 as follows:

"2/ A Chemical analysis shall be performed on each lot."

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* Delete paragraph 4.5.1.2 in its entirety.

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* Add new paragraph 4.5.1.2 as follows:

"4.5.1.2 Powder metal preform. The test specimens for a warhead lot that are to be machined from powder metal preforms shall be obtained from one of the following methods.

a. All specimens from the first method are obtained from the sidewall of the powder metal preform. The test specimens shall be produced to such dimensions so that yield and impact test bars shall be machined to the form and dimensions shown in figures 4 and 5. Two test specimens of each type shall be machined from each body.

b. The yield specimens shall be obtained from the sidewall while the impact specimens shall be obtained from the base of the powder metal preform. The test specimens (shall be produced of such dimensions so that the yield and impact test bars) shall be machined to the form and dimensions shown in figures 4 and 5. Two test specimens of each type shall be machined from each body."

* Delete paragraph 4.5.2 in its entirety.

* Add new paragraph 4.5.2 as follows:

"4.5.2 Impact resistance of warhead body.

4.5.2.1 Forged Warhead body. Charpy v-notch impact test shall be conducted in accordance with ASTM E23. The temperature of the test specimen at the time of the fracture shall be -40 ±2F. The error in the Charpy Machine shall not exceed ±1 ft - lb, when the machine is tested using comparison specimens prepared by U.S. Army materials Research Agency (see 6.6). This comparison shall have been made within one year prior to the time of the inspection testing.

4.5.2.2 Powder metal preform. The temperature of the test specimen at the time of the fracture shall be -40 ±2F. The error in the Charpy machine shall not exceed ±1 ft-lb, when the machine is tested using comparison specimens prepared by U.S. Army Materials Research Agency (see 6.6). This comparison shall have been made within one year prior to the time of the inspection testing.

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* 4.5.3.2 Powder metal preform: Delete "Tension tests shall be accomplished in accordance with ASTM E8, except that the rate of strain will be 0.003 to 0.007 inch per inch per minute through the yield strength and then is increased so as to produce failure in approximately one additional minute." and substitute "Tension tests shall be accomplished in accordance with ASTM E8, except that the configuration shall be as per figure 5, and the rate of strain will be 0.003 to 0.007 inch per inch per minute through the yield strength and then to be increased so as to produce failure in approximately one additional minute."

4.5.8 add the following:

"The HTS circuit isolation shall be tested at 200 VDC with a pressure of 10 PSI applied to the HTS shield."

* Add new paragraph 4.5.10 as follows:

"4.5.10 Chemical analysis (Powder metal only). A chemical analysis shall be performed on each lot of preforms produced and should identify all elements shown to be crucial by the drawing. The chemical analysis shall be performed in accordance with Method 111 or Method 112 of FED-STD-151."

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* Delete "Figure 1. Location of test specimens in warhead body" and substitute "Figure 1. Location of test specimens in forged warhead body"

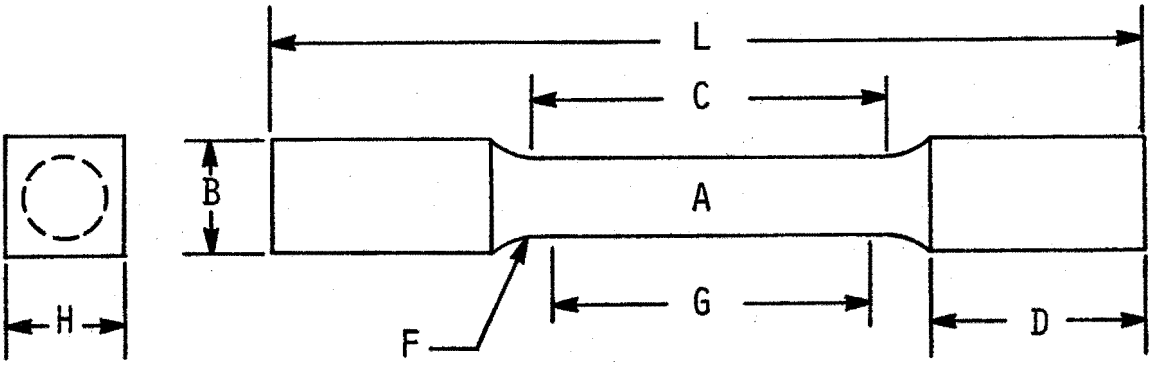
* Add new Page 23A.

* Add new Page 23B.

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* 6.3 Submission of inspection equipment designs for approval: Delete",Dover, NJ 07801." and substitute the following "(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirement List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply."

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ALL DIMENSIONS IN INCHES

DIMENSIONS	
A- DIAMETER AT CENTER[1]	0.157 ± 0.001
B- WIDTH OF GRIP ENDS APPROX.[3]	0.250
C- LENGTH OF REDUCED SECTION MIN.	1.00
D- GRIP LENGTH, APPROX.[3]	1.250
F- FILLET RADIUS, MIN.	0.250
G- GAGE LENGTH	0.750 ± 0.005
H- DEPTH OF GRIP ENDS APPROX.	0.250
L- TOTAL LENGTH, APPROX.,[2,3]	4.00
AREA OF CROSS SECTION SQUARE INCHES APPROX.	1/2

1. Diameter of the reduced section may be smaller at the center than at ends. Difference shall not exceed 1 percent of the diameter at the center.

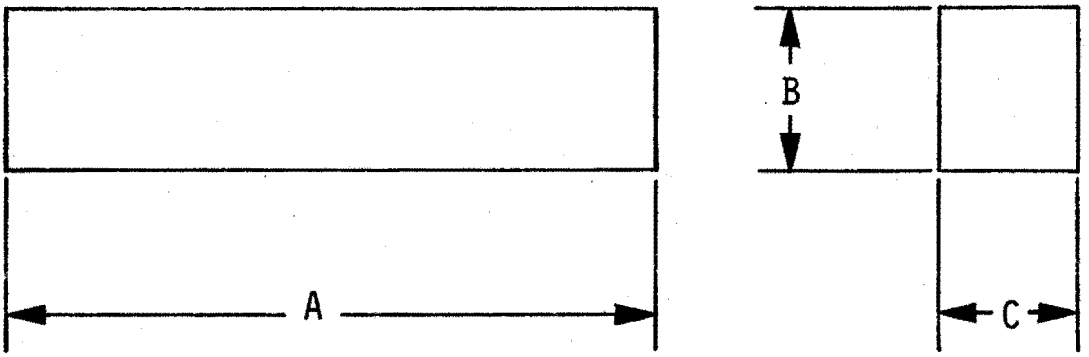
2. When extensometer is required to determine elastic properties, dimensions C and L may be modified to provide therefor. In all cases the percentage of elongation shall be based on dimension G.

3. Ends may be any shape to fit the holders of the testing machine in such a manner that the lead in axial.

4. A generous fillet is desirable.

FIGURE 4. Powder metal preform tensile specimen

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DIMENSION	MM	TOLERANCE + INCHES -		INCHES	TOLERANCE + INCHES -	
A- LENGHT	55.000	0.0	2.50	2.165	0.0	0.100
B- WIDTH	3.912	0.025	0.025	0.154	0.001	0.001
C- DEPTH	3.912	0.025	0.025	0.154	0.001	0.001

FIGURE 5. Powder metal preform impact specimen

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* Delete paragraph 6.4 in its entirety, and subsitute the following:

"6.4 Process control document. The process control document shall be submitted in accordance with the process control document requirements of data item description DI-R-5297A (Tailored), to Commander, U.S. Army Armament Munitions and Chemical Command, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000 prior to the start of production. Questions, if any, regarding proprietary issues of the process control documentation may be directed to the command legal office ATTN: AMSMC-GC at the above address."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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